

3 Failure: Corrosion and oxidation

1. Simple definition of oxidation and corrosion

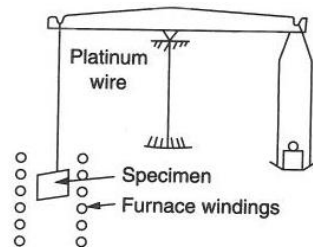
Oxidation: removal of electrons to create positive ions,
e.g. $\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^-$;

Dry Corrosion: chemical reaction of a dry metal surface with
an oxidising gas;

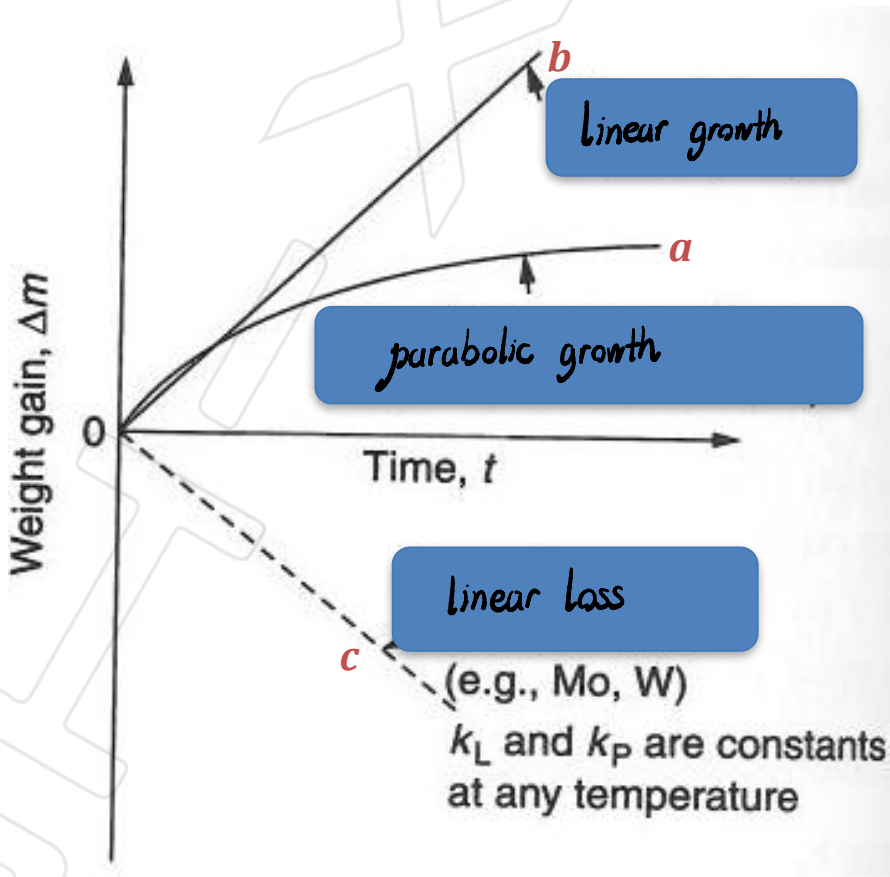
Wet Corrosion – chemical attack through water.

2. Oxidation kinetics

It is common to measure kinetics by the change in mass versus
time:



There are three types of oxidation kinetics as shown in the figure below.



- a is parabolic growth ($\Delta m^2 = k_p t$), for which the oxidation rate is dependent on the time and the material will show weight gain.
- b is linear growth ($\Delta m = k_l t$), for which the oxidation rate is independent of the time and the material will show weight gain.
- c is linear loss ($\Delta m = -k_l t$), for which the oxidation rate is independent of the time and the material will show weight loss.

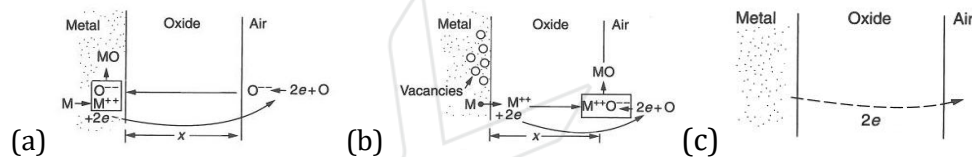
$$k_p = A_p e^{-\frac{Q_p}{RT}}$$

$$k_l = A_l e^{-\frac{Q_l}{RT}}$$

The kinetic constants (k_p and k_l) follow an Arrhenius law so the rate of oxidation increase exponentially with temperature increase.

3. Micromechanisms of oxidation. ($M^{2+} + 2e^- + O^{2-} = MO + 2e^-$)

(1) For parabolic corrosion



- (i) When O^{2-} diffuse very slowly (e.g. Fe), M^{2+} and two electrons diffuse outwards through the oxide to meet the O^{2-} at the surface. This is the situation for Figure b.
- (ii) When M^{2+} diffuses very slowly. E.g. Ti, the O^{2-} diffuse inwards through the oxide to meet the M^{2+} at the metal and oxide interface. This is Figure a.
- (iii) When e^- move slowly in oxide. E.g. Al, the oxide growth is limited by the motion of electrons through the oxide. Slow growth at either interface. This is Figure c.

For parabolic corrosion, a protective oxide is one that forms a stable oxide layer that grows very slowly. If the diffusion coefficient is low, the layer thickness will remain thin for a long time, such as Al_2O_3 on Al.

$$x^2 = k_p t$$

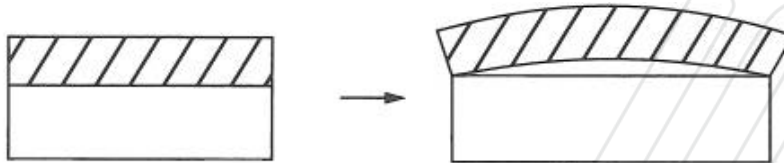
Where: $k_p \propto -2D_0 e^{-\frac{Q}{RT} \Delta C}$

(2) For linear growth

Linear growth occurs when the oxide layer breaks-up or spalls. If the density of oxide > density of metal, the oxide splits open from metal as in image below.



If the density of oxide $\leftarrow$ density of metal, the oxide shrinks away from the metal as shown below.



In either case, the oxide does not form a barriers to oxygen reacting with the metal and growth is independent of the oxide thickness.

(3) For linear loss

Linear loss in mass occurs when the oxide is volatile. During high temperature oxidation of W or Mo, oxide evaporates and forms no layer. The rate of oxidation is then independent of time.

3. Application: Ni-based turbine blade

A quick calculation according to the slides 18-19:

From the table on Slide 18, for Ni to be oxidized to a depth of 0.1 mm at 0.7 T_m in air is 600 hours. If the blade is oxidized for 5000 hours, what is the thickness of the Ni oxides?

$$x^2 = k_p t$$

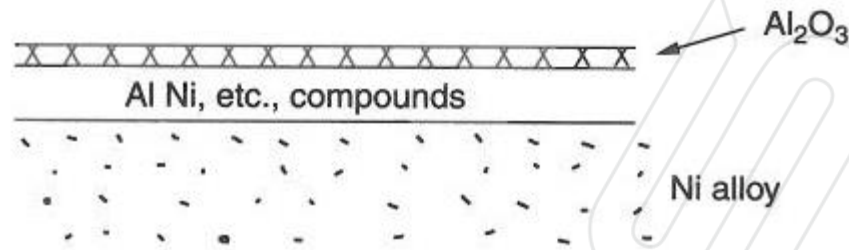
$$k_p = \frac{x^2}{t}$$

$$\therefore x_{10}^2 = \frac{x_0^2}{t_0} t_1 = 0.289 \text{ (mm)}$$

which cannot be accepted for use in aero-engines.

Therefore Cr is added to Ni alloys to increase the oxidation resistance. Chromium has a much higher affinity to oxygen than Ni (high driving force). From kinetic point of view, Chromium needs 1600 hours (read from the table on Slide 18) to be oxidized to a depth of 0.1 mm at 0.7 T_m in air, which is much longer compared to Ni (600 hours). When Cr is used at the homologous T/T_m of Ni, the time for such an oxidation is even longer, indicating Cr is a very good alloying element against oxidation/corrosion for Ni alloys.

Another way to improve oxidation resistance of Ni alloys is coating with Al. Spray droplets of Al onto the surface of the blade to a few μm thick. Heat blade in furnace so some Al diffuses in to form AlNi and the rest oxidises to an Al_2O_3 layer, which is a good isolator to prevent fast oxidation of Ni.



5. Disadvantages of oxide films for material mechanical properties

- Oxides are brittle. They can form cracks, especially when the temperature of the blade changes. They can crack when differential thermal expansion coefficients between alloy and oxide lead to stresses.
- Oxide layers act as crack initiation sites for thermal fatigue cracks and the crack can spread into the alloy itself.

6. Influence factors for oxidation rate?

temperature, diffusion coefficient, morphology of oxygen layer ...